Setup

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1 Setting Up Your Python Development Environment

CMIT Summer School 2026
A Step-by-Step Guide for Windows 11 and macOS

1.1 Overview

This guide will walk you through: 1. **Installing Conda** (via Miniconda or Anaconda) on Windows 11 and macOS. 2. **Creating and managing isolated Python environments** using `conda`. 3. **Installing essential packages** like `numpy` and `pytorch`. 4. **Integrating your Conda environment with VS Code** for a smooth development experience.

> **Why Conda?** Conda is a powerful package and environment manager that runs on Windows, macOS, and Linux.[reference:0] It allows you to create separate environments for different projects, each with its own Python version and packages, preventing dependency conflicts.[reference:1]
>
> **Miniconda vs. Anaconda**: Miniconda is a lightweight version that includes only `conda` and its dependencies. Anaconda includes Miniconda plus a large collection of pre-installed scientific packages (~3 GB).[reference:2] For most users, **Miniconda is recommended** as it's faster to download and you can install only what you need.[reference:3]

1.2 Part 1: Installing Conda

1.2.1 Windows 11

Option A: Using `winget` (Recommended – Fastest)

1. Open **PowerShell** or **Command Prompt** as **Administrator** (right-click → "Run as administrator").
2. Run the following command:

   ```
   winget install -e --id Anaconda.Miniconda3
   ```

   This will download and install Miniconda automatically.[reference:4]

Option B: Manual Installation (Graphical)

1. Visit the Miniconda download page or the Anaconda download page.
2. Download the **Windows 64-bit installer** (e.g., `Miniconda3-latest-Windows-x86_64.exe`).[reference:5]
3. **Double-click the `.exe` file** to launch the installer.[reference:6][reference:7]
4. Follow the on-screen instructions. **If unsure about any setting, accept the defaults.**[reference:8]
5. **Important**: During installation, you may be asked whether to "Add Miniconda3 to my PATH environment variable." **Check this box** – it makes it easier to use `conda` from any terminal.[reference:9]

Verify Installation (Both Options)

1. Open a **new** Command Prompt or PowerShell window.
2. Run:

   ```
   conda --version
   ```

   You should see something like `conda 24.x.x`.

3. To see a list of installed packages, run:

   ```
   conda list
   ```

   A list of packages will appear if Conda is installed correctly.[reference:10]

   Tip: On Windows, you can also open the **Anaconda Prompt** from the Start Menu, which automatically configures the environment for Conda.[reference:11]

1.2.2 macOS

> **Important Note for Apple Silicon (M1/M2/M3) Users**: As of August 2025, Anaconda has stopped building packages for Intel Mac computers (`osx-64`).[reference:12] Make sure to download the **Apple Silicon** version if you have an M-series chip.[reference:13]

Option A: Graphical Installer (Recommended)

1. Visit the Miniconda download page or the Anaconda download page.
2. Download the **macOS graphical installer** (`.pkg` file) – choose either **Intel** or **Apple Silicon** based on your Mac.[reference:14]
3. **Double-click the `.pkg` file** to launch the installer.[reference:15]
4. Follow the prompts on the installer screens. **If unsure about any setting, accept the defaults.**[reference:16]

Option B: Command-line Installer (Terminal)

1. Open **Terminal** (you can find it via Spotlight search – press `Cmd + Space` and type "Terminal").[reference:17]
2. Download the installer script:

   ```
   curl -O https://repo.anaconda.com/miniconda/Miniconda3-latest-MacOSX-x86_64.sh
   ```

(For Apple Silicon, replace `x86_64` with `arm64` in the URL.)

3. Run the installer:

```
bash Miniconda3-latest-MacOSX-x86_64.sh
```

4. Follow the prompts. When asked **"Do you wish the installer to initialize Miniconda3 by running conda init?"**, answer `yes` (recommended).[reference:18]

5. **Close and re-open your terminal** for the changes to take effect.[reference:19]

Verify Installation

1. In a **new** terminal window, run:

```
conda --version
```

You should see the Conda version number.

2. To see a list of installed packages, run:

```
conda list
```

A list of packages will appear if Conda is installed correctly.[reference:20]

1.3 Part 2: Creating and Managing Conda Environments

Conda allows you to create isolated environments for different projects.[reference:21] This is essential to avoid dependency conflicts.

Basically when Python complains something is missing (eg torch), we need to (1) Create cmit_env once and (2) install required or missing packages on cmit_env through "command window" of anaconda.

1.3.1 2.1 Create a New Environment

The basic command to create an environment is:[reference:22]

```
conda create -n <env-name> python=<version> [packages...]
```

Example: Create an environment called `cmit_env` with Python 3.11 and install `numpy` and `pandas` at the same time:[reference:23]

```
conda create -n cmit_env python=3.11 numpy pandas
```

> Conda will ask you to confirm the installation. Type `y` and press Enter to proceed.[reference:24]

1.3.2 2.2 Activate the Environment

Once created, you need to **activate** the environment to use it:

```
conda activate cmit_env
```

You'll notice the environment name appears in your terminal prompt, e.g., `(cmit_env) C:\>` or `(cmit_env) $`.

1.3.3 2.3 List All Environments

To see all your environments (the active one is marked with *):[reference:25]

```
conda info --envs
```

1.3.4 2.4 Deactivate the Environment

To return to the `base` environment:

```
conda deactivate
```

1.4 Part 3: Installing Packages with Conda

Once your environment is activated, you can install packages using `conda install`.

1.4.1 3.1 Install NumPy

```
conda install numpy
```

1.4.2 3.2 Install PyTorch (CPU Version – Recommended for Beginners)

For a CPU-only version (works on any computer, no GPU required):[reference:26]

```
conda install pytorch torchvision torchaudio cpuonly -c pytorch
```

1.4.3 3.3 Install PyTorch (GPU Version – If you have an NVIDIA GPU)

If you have a CUDA-capable NVIDIA GPU, you can install the GPU version:[reference:27]

```
conda install pytorch torchvision torchaudio -c pytorch
```

> **Tip**: To find the exact command for your system, visit the PyTorch official website and use their installer configuration tool.

1.4.4 3.4 Install Multiple Packages at Once

You can install several packages in one command:

```
conda install numpy matplotlib jupyter notebook scipy
```

1.4.5 3.5 Install a Package Without Activating the Environment

You can also specify the environment name directly:[reference:28]

```
conda install --name cmit_env numpy
```

1.5 Part 4: Integrating Conda with VS Code

VS Code has excellent support for Conda environments.[reference:29] Here's how to set it up.

1.5.1 4.1 Install VS Code

If you haven't already: 1. Visit code.visualstudio.com. 2. Download the installer for your operating system. 3. Run the installer and follow the instructions.

1.5.2 4.2 Install the Python Extension

1. Open VS Code.
2. Click the **Extensions** icon in the left sidebar (or press `Ctrl+Shift+X` / `Cmd+Shift+X`).
3. Search for **"Python"** (by Microsoft).
4. Click **Install**.

1.5.3 4.3 Select Your Conda Environment as the Python Interpreter

Method 1: Using the Command Palette (Quick Switch)

1. Open your project folder in VS Code (`File > Open Folder`).
2. Open the **Command Palette** by pressing `Ctrl+Shift+P` (Windows) or `Cmd+Shift+P` (macOS).[reference:30]
3. Type and select **`Python: Select Interpreter`**.[reference:31]
4. From the list, choose the interpreter that corresponds to your Conda environment (e.g., `Python 3.11.0 ('cmit_env': conda)`).[reference:32]

Method 2: Click the Interpreter in the Status Bar

1. Look at the **bottom-left corner** of the VS Code window – you'll see the current Python interpreter (e.g., `Python 3.11.0`).
2. Click on it.
3. A list of available interpreters will appear – select your Conda environment.

1.5.4 4.4 Verify the Environment is Active

1. Open a new terminal in VS Code (`Terminal > New Terminal` or ``Ctrl+` ``).

2. The terminal prompt should show your environment name in parentheses, e.g., `(cmit_env) C:\...` or `(cmit_env) $`.

3. If it doesn't automatically activate, you can manually run:[reference:33]

   ```
   conda activate cmit_env
   ```

1.5.5 4.5 (Optional) Pin the Environment for a Project

To automatically use the correct environment every time you open a specific project folder, you can create a `.vscode/settings.json` file inside your project:[reference:34]

```
{
    "python.defaultInterpreterPath": "C:\\Users\\<YourUsername>\\miniconda3\\envs\\cmit_env\\py
}
```

   On macOS, the path would be something like: `/Users/<YourUsername>/miniconda3/envs/cmit_env/bin/python`

1.5.6 4.6 Using Conda Environments in Jupyter Notebooks (VS Code)

If you want to use your Conda environment as a kernel in Jupyter Notebooks within VS Code:[reference:35]

1. Activate your environment:

```
conda activate cmit_env
```

2. Install `ipykernel`:

```
conda install ipykernel
```

3. Register the kernel:

```
python -m ipykernel install --user --name cmit_env --display-name "Python (cmit_env)"
```

4. In VS Code, when you open a `.ipynb` file, click on the **Kernel** button in the top-right corner and select `Python (cmit_env)` from the list.

1.6 Quick Reference: Essential Commands

Task	Command
Check Conda version	conda --version
List all environments	conda info --envs
Create new environment	conda create -n <name> python=<version>
Create environment with packages	conda create -n <name> python=<version> numpy pandas
Activate environment	conda activate <name>
Deactivate environment	conda deactivate
Install package (active env)	conda install <package>
Install package (specific env)	conda install --name <env> <package>
Install PyTorch (CPU)	conda install pytorch torchvision torchaudio cpuonly -c pytorch
Install PyTorch (GPU)	conda install pytorch torchvision torchaudio -c pytorch
List installed packages	conda list
Remove environment	conda env remove -n <name>

1.7 Troubleshooting

1.7.1 `conda` command not found

- **Windows**: Make sure you checked "Add Miniconda3 to my PATH" during installation. If not, reinstall or manually add the Conda `Scripts` folder to your PATH.[reference:36]
- **macOS**: Close and re-open your terminal. If still not found, run `source ~/.zshrc` (or `~/.bash_profile`) to reload your shell configuration.[reference:37]

1.7.2 VS Code doesn't show my Conda environment

1. Make sure the **Python extension** is installed.
2. Try **reloading the VS Code window** (`Ctrl+Shift+P` → `Developer: Reload Window`).
3. If still not showing, you can manually specify the Conda environment path in VS Code settings:
   - Open settings (`Ctrl+,` / `Cmd+,`).
   - Search for `python.condaPath` and set it to the path of your Conda executable (e.g., `C:\Users\<User>\miniconda3\Scripts\conda.exe` on Windows).[reference:38]

1.7.3 SSL/Certificate errors (Windows)

If you encounter SSL errors when installing packages (like the one we saw in Lecture 1), try:

```
conda install openssl=3.5.6
```

1.8 Summary

You have now set up: - Conda (Miniconda or Anaconda) on your system. - A dedicated Python environment (`cmit_env`) with `numpy` and `pytorch`. - VS Code configured to use your Conda environment.

You're now ready to start the CMIT Summer School lectures!